



**JAWAHARLAL COLLEGE
OF ENGINEERING & TECHNOLOGY**
JAWAHAR GARDENS, LAKKIDI MANGALAM, PALAKKAD
(An ISO 9001:2015 Certified Institution)



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (CYBERSECURITY)

ESP32-BASED DEAUTHENTICATION ATTACK DETECTION SYSTEM

ABSTRACT

Wi-Fi deauthentication attacks are a common wireless security threat in which an attacker sends forged deauthentication packets to disconnect devices from a Wi-Fi network. This project proposes an ESP32-based Deauthentication Attack Detection System designed to monitor wireless traffic and identify suspicious deauthentication activity in real time. The ESP32 continuously observes Wi-Fi packets and analyzes deauthentication frames to detect unusual or excessive packet activity that may indicate an ongoing attack. When a potential deauthentication attack is detected, the system provides an immediate alert through the ESP32, allowing users to recognize the threat quickly. The system is designed to be low-cost, portable, and suitable for wireless security monitoring and educational cybersecurity applications. By providing real-time detection and notification, the proposed system helps improve awareness of Wi-Fi attacks and can serve as a foundation for developing more advanced wireless intrusion detection systems.

Submitted by,

AKSHAY N, JCE23CC008

Recommendation by Guide

Ms. FARBEENA N, Assistant Professor

Dept. of Computer Science

JCET, LAKKIDI, KOOTUPATHA

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